A 72-VERTEX COUNTEREXAMPLE TO A SPECTRAL-GAP CONJECTURE OF GREAVES AND ZHU

Abstract. Let $P_m(n)$ be the generalised pancake graph on the colored permutation group $\mathbb{Z}_m \wr S_n$ and let $\mathcal{P}_{m,n}$ be its colour–position equitable partition, with quotient matrix $Q(\mathcal{P}_{m,n})$. Conjecture 2 of Greaves and Zhu asserts that the two spectral gaps agree, $\lambda_1(P_m(n)) - \lambda_2(P_m(n)) = \lambda_1(Q(\mathcal{P}_{m,n})) - \lambda_2(Q(\mathcal{P}_{m,n}))$, for all $m \geq 1$ and $n \geq 2$. It is false at $(m,n) = (6,2)$. On $P_6(2)$, a 4-regular graph on 72 vertices, the integer vector taking the values $2, 1, -1, -2, -1, 1$ on the six classes of the colour difference sums to zero and satisfies $Av = 3v$, so the graph's gap is at most 1, whereas $Q(\mathcal{P}_{6,2})$ has second eigenvalue $2\sqrt{2}$ and gap $4-2\sqrt{2} > 1$. We claim no counterexample of least order.

1. The conjecture

Fix integers $m \geq 1$ and $n \geq 2$. A vertex of the *generalised pancake graph* $P_m(n) = \mathrm{Cay}(\mathbb{Z}_m \wr S_n, R_m)$ is a pair (w, c) in which $w = w_1 \cdots w_n$ is a permutation word of $[n]$ — position j carries the letter w_j — and $c = (c_1, \ldots, c_n) \in \mathbb{Z}_m^n$ is a colour vector, position j carrying the colour c_j. For $k \in [n]$ and $\varepsilon \in \{+1, -1\}$ the generator r_k^ε reverses the prefix of length k and adds ε to the colour of every entry that sat in positions $1, \ldots, k$:

$$r_k^\varepsilon(w,c) = (w', c'), \qquad \begin{cases} w'_j = w_{k+1-j},\ c'_j = c_{k+1-j} + \varepsilon, & j \leq k, \\ w'_j = w_j,\ c'_j = c_j, & j > k. \end{cases} \tag{1}$$

For $m \geq 3$ one takes $R_m = \{r_k^+, r_k^- : k \in [n]\}$, so that $P_m(n)$ has $m^n n!$ vertices and is $2n$-regular. Only the cell $m = 6$ is used below, so the separate convention at $m \leq 2$ — a smaller connection set — is never invoked, and nothing here applies to $m \leq 2$. Write A for the adjacency matrix and $\lambda_1 \geq \lambda_2 \geq \cdots$ for its eigenvalues.

The *colour–position partition* $\mathcal{P}_{m,n}$ has the mn cells

$$V_{i,j} = \big\{(w,c) : \text{the letter 1 sits at position } j \text{ with colour } i\big\}, \tag{2}$$

one cell for each $i \in \mathbb{Z}_m$ and $j \in [n]$. It is equitable, and with the cells ordered lexicographically by (i,j) its quotient matrix is the block circulant

$$Q(\mathcal{P}_{m,n}) = \mathrm{circ}(2D_n, E_n, O, \ldots, O, E_n), \tag{3}$$

where $D_n = \mathrm{diag}(0, 1, \ldots, n-1)$ and $(E_n)_{ij} = [\, i + j \leq n+1\,]$. Set

$$\psi_{m,n} = \lambda_1(A) - \lambda_2(A), \qquad \beta_{m,n} = \lambda_1\big(Q(\mathcal{P}_{m,n})\big) - \lambda_2\big(Q(\mathcal{P}_{m,n})\big).$$

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The generator rule (1), the partition (2) and the ordering of its cells are those of [3]: the connection set is their R_m for $m \geq 3$, the cells are indexed by the colour and the position of the letter 1, and (3) is the block-circulant quotient that [3] print for that partition. As a check on this agreement, the accompanying program builds $P_6(2)$ from (1) alone, forms the partition (2), and recovers (3) from the resulting graph rather than transcribing it.

The statement we refute is [3, Conjecture 2], quoted here verbatim from the source of arXiv:2509.09425v2:

> Let $m \geq 1$ and $n \geq 2$ be integers. Then $\lambda_1(P_m(n)) - \lambda_2(P_m(n)) = \lambda_1(Q(\mathfrak{P}_{m,n})) - \lambda_2(Q(\mathfrak{P}_{m,n}))$.

That is, $\psi_{m,n} = \beta_{m,n}$ for all $m \geq 1$, $n \geq 2$. The same source records that for $m \geq 2$ the conjecture is open (the case $m = 1$ is a known theorem on initial-reversal Cayley graphs; see [3, 1]); it restates a conjecture of Blanco and Buehrle [2], and Blanco [1] proves the inequality $\psi_{m,n} \leq \beta_{m,n}$ while calling the equality open. Since the conjecture is universally quantified, one cell decides it. The label "Conjecture 2" is that of arXiv:2509.09425v2; the printed journal version may number it differently, but the statement is reproduced above, so what is refuted does not depend on the label.

2. The colour difference

Lemma 1. *Let $m \geq 3$ and $n = 2$. For a vertex σ of $P_m(2)$ let $d(\sigma) \in \mathbb{Z}_m$ be the colour of the letter 1 minus the colour of the letter 2. Then the four neighbours of σ carry colour differences $d+1$, $d-1$, d, d.*

Proof. The letters 1 and 2 occupy positions 1 and 2 in one order or the other. By (1), r_1^ε adds ε to the colour in position 1 and leaves position 2 untouched, so it changes d by $+\varepsilon$ if the letter 1 is in position 1 and by $-\varepsilon$ if the letter 2 is; either way the unordered pair $\{r_1^+\sigma, r_1^-\sigma\}$ realises the differences $d+1$ and $d-1$. And r_2^ε reverses both positions and adds ε to both colours, so both letters' colours move by ε and d is unchanged. $\square$

3. The cell $(6, 2)$: 72 vertices

Theorem 2. *Let $H = P_6(2)$, a connected simple 4-regular graph on 72 vertices. Then*

$$\psi_{6,2} \leq 1 < 4 - 2\sqrt{2} = \beta_{6,2} = 1.1715728752538099\ldots,$$

a strict deficit of at least $3 - 2\sqrt{2} = 0.1715728752538099\ldots$. In particular [3, Conjecture 2] *is false.*

Proof. The graph's side. Let v be the integer vector depending only on the colour difference $d \in \mathbb{Z}_6$ through the table

(4)

d	0	1	2	3	4	5
v	2	1	-1	-2	-1	1

that is, $v = 2\cos(\pi d/3)$. Each of the six classes holds 12 of the 72 vertices, so $\sum v = 12(2+1-1-2-1+1) = 0$ and $v \perp \mathbf{1}$; and $v \neq 0$. By Lemma 1 the four neighbours of a vertex with difference d have differences $d+1, d-1, d, d$, so

$$(Av)(\sigma) = v(d+1) + v(d-1) + 2v(d) = 1 \cdot v(d) + 2v(d) = 3v(d),$$

using $v(d+1) + v(d-1) = 2\cos(\pi/3)\, v(d) = v(d)$. Thus $Av = 3v$ exactly over $\mathbb{Z}$. One row in full: at $w = 12$, $c = (0, 0)$ we have $d = 0$ and $v = 2$, the four neighbours are $r_1^+ = (12, (1, 0))$, $r_1^- = (12, (5, 0))$, $r_2^+ = (21, (1, 1))$, $r_2^- = (21, (5, 5))$ with differences $1, 5, 0, 0$ and values $1, 1, 2, 2$, and $1 + 1 + 2 + 2 = 6 = 3 \cdot 2$. Since H is connected and 4-regular, $\lambda_1 = 4$ is simple, so $\lambda_2 \geq 3$ and $\psi_{6,2} \leq 1$.

The quotient's side. By (3), $Q(\mathcal{P}_{6,2}) = \operatorname{circ}(2D_2, E_2, O, O, O, E_2)$ with $2D_2 = \operatorname{diag}(0, 2)$ and $E_2 = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. This 12×12 matrix is $I_6 \otimes 2D_2 + C_6 \otimes E_2$ with C_6 the adjacency matrix of the 6-cycle; diagonalising C_6 makes it orthogonally similar to $\bigoplus_{r=0}^{5} M_r$ with $M_r = 2D_2 + 2\cos(2\pi r/6)E_2$, and the four distinct blocks are integral:

$$M_0 = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}, \qquad M_1, M_5 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix},$$

$$M_2, M_4 = \begin{pmatrix} -1 & -1 \\ -1 & 2 \end{pmatrix}, \qquad M_3 = \begin{pmatrix} -2 & -2 \\ -2 & 2 \end{pmatrix},$$

so that

$$\det\bigl(xI - Q(\mathcal{P}_{6,2})\bigr) = x(x-4)\,(x^2-8)\,(x^2-3x+1)^2\,(x^2-x-3)^2.$$

Hence $\lambda_1(Q) = 4$, and $\lambda_2(Q) = 2\sqrt{2}$: it is a root of $x^2 - 8$, and each other block polynomial g satisfies $g(2\sqrt{2}) > 0$ with $g'(2\sqrt{2}) > 0$, namely $9 - 6\sqrt{2} > 0$ (as $81 > 72$) and $5 - 2\sqrt{2} > 0$, so neither has a root at or above $2\sqrt{2}$; and the remaining root of M_0 is $0 < 2\sqrt{2}$. Therefore $\beta_{6,2} = 4 - 2\sqrt{2}$.

The comparison. $4 - 2\sqrt{2} > 1$ is $3 > 2\sqrt{2}$, i.e. $9 > 8$; equivalently $x^2 - 8$ takes the value $1 > 0$ at $x = 3$. So $\psi_{6,2} \leq 1 < \beta_{6,2}$. □

4. Scope

What is refuted. [3, Conjecture 2] as a universally quantified statement over all $m \geq 1$ and all $n \geq 2$. One cell settles that, and the cell exhibited is $(6, 2)$ on 72 vertices.

What is not settled. No other cell. The argument above bounds $\psi_{6,2}$ above and $\beta_{6,2}$ below at $m = 6$, $n = 2$ only; nothing here bears on any $m \neq 6$ at $n = 2$, on the cells $m \leq 2$ — which use a connection set other than the R_m of Section 1, nowhere built or analysed here — or on any cell with $n \geq 3$. In particular we do *not* claim that $(6, 2)$ is a counterexample of least order. The inequality $\psi_{m,n} \leq \beta_{m,n}$ of [1] is a theorem and is untouched; so is the $m = 1$ case.

5. Verification

The accompanying program `verify.py` builds $P_6(2)$ vertex by vertex from (1) and checks that it is simple, 4-regular, connected and has 72 vertices; it rebuilds the partition (2) and verifies that it is equitable by comparing the neighbour-count vector of *every* vertex of a cell, then checks that its quotient is the block circulant (3); it verifies $Av = 3v$ at all 72 vertices in exact integer arithmetic; it recomputes $\det(xI - Q(\mathcal{P}_{6,2}))$ by Faddeev–LeVerrier over $\mathbb{Q}$; and it pins $\lambda_2(Q) = 2\sqrt{2}$ by exact eigenvalue *counts* — Sylvester's law of inertia applied to $tI - Q$ for rational t — rather than by a numerical eigensolver. No floating-point number takes part in any decision, and no third-party package is used. Its recorded run is `verify.output.txt`; that run also covers cells and identities beyond the one used here, which this note does not rely on.

Two limits of the program are worth stating. It never computes $\lambda_2(A(P_6(2)))$, only bounds it below by the eigenvector (4), which is all that $\psi_{6,2} \le 1$ needs; and it decides nothing about minimality.

References

[1] S. A. Blanco, *On the Laplacian spectral gap of generalized pancake graphs*, arXiv:2608.15398v1, 2026. arXiv:2608.15398.

[2] S. A. Blanco and C. Buehrle, *On the spectra of prefix-reversal graphs*, arXiv:2506.08345, 2025. arXiv:2506.08345.

[3] G. Greaves and H. Zhu, *A note on some spectral properties of generalised pancake graphs*, Discrete Math. **349** (2026), no. 9, 115102. doi:10.1016/j.disc.2026.115102; arXiv:2509.09425v2.